

Solutions — Randomization Tests

Randomization Tests

*Statistical inference is primarily concerned with understanding and quantifying the uncertainty of parameter estimates. While the equations and details change depending on the setting, the foundations for inference are the same throughout all of statistics. In this chapter, we introduce the **hypothesis testing framework** — a formal method for evaluating competing claims about a population using data. We learn how to build **randomization distributions** by shuffling data under the assumption that there is no effect, and we use these distributions to compute **p-values** that quantify the strength of the evidence.*

KEY CONCEPTS

- **Set up a test.** Recognize when a statistical test is needed, and state null and alternative hypotheses in terms of a parameter.
- **Model chance with a randomization distribution** built under the assumption that the null hypothesis is true.
- **Find and interpret a p -value** from that distribution — including one- versus two-tailed tests — as the strength of evidence against the null hypothesis.
- **Make and state a decision** by comparing the p -value to a significance level, then conclude in context.
- **Know what a test can and cannot tell you:** Type I and Type II errors, the significance level as the tolerable Type I error rate, and why statistical discernibility is not the same as practical importance.

Motivation: Cereal example

ACTIVITY

Consider the following scenario: A sample of 80 four-to six-year olds participated in a study in which they were asked to taste two cereals, both called “Oat Bits” (in random order). The cereal in each box was exactly the same, but the boxes differed in the fact that one of the boxes had pictures of a star mascot and the other did not.



Figure 1: The two cereal boxes shown to the children. The cereal inside was identical; only one box featured a cartoon star mascot.

- a) If the cartoons make no difference, because after all the cereal is the same in each box, how many of the 80 children do you expect to pick the cartoon box? What sample proportion is this?

SOLUTION

1. If the cartoons make no difference, each child is equally likely to pick either box.
2. We expect about half: 40 of the 80 children.
3. As a proportion, $\hat{p} = 40/80 = 0.50$.

- b) Suppose that in this study, 52 of the children chose the box with cartoons. What sample proportion is this? Does this lead you to believe that there was a preference toward the cartoon box, or would you expect this sample proportion to occur by chance even if there was no preference? (We will use StatLens to explore the sampling distribution for the proportion to determine what values are likely to occur.)

SOLUTION

1. $\hat{p} = 52/80 = 0.65$, which is above the 0.50 we would expect if there were no preference.
2. On its own, 0.65 is not proof: sample proportions vary from sample to sample even when the true proportion is 0.50.
3. The real question is whether a value as high as 0.65 is *plausible by chance alone*. To answer it, we simulate many samples of 80 children under “no preference” ($p = 0.50$) and see how often chance produces a proportion this large.

- c) Using the simulation from part b), what sample proportions do we expect to see if there is no preference toward the cartoons?

SOLUTION

1. Under “no preference” ($p = 0.50$), the simulated sample proportions cluster around 0.50.
2. Almost all fall roughly between 0.39 and 0.61 (about 31 to 49 of the 80 children).
3. The observed $\hat{p} = 0.65$ sits beyond that typical range, in the tail — so chance alone is an unlikely explanation. We will make this precise with a p -value later in the chapter.

The hypothesis testing framework

There is almost always variability in data — one dataset will not be identical to a second dataset even if they are both collected from the same population using the same methods. However, quantifying the variability in the data is neither obvious nor easy. Answering the question “*how different is one dataset from another?*” is not trivial.

The question at the heart of hypothesis testing is: *Could the pattern we observe in the data have arisen from chance alone, or is there something real going on?*

Hypothesis testing

A *hypothesis test*, or *statistical test* uses data from a sample to assess a claim about a population.

A *hypothesis test* is a statistical technique used to evaluate competing claims using data. The process involves:

1. *State the hypotheses*: Define a null hypothesis (H_0) and an alternative hypothesis (H_A).
2. *Collect data*: Gather evidence that bears on the hypotheses.
3. *Assess the evidence*: Determine how likely the observed data would be if the null hypothesis were true.
4. *Draw a conclusion*: Decide whether the evidence is strong enough to reject the null hypothesis.

If the null hypothesis and the data notably disagree, then we reject the null hypothesis in favor of the alternative hypothesis. There are many nuances to hypothesis testing, and we will discuss these ideas throughout this chapter and those that follow.

Null and alternative hypotheses

Usually, a *null hypothesis*, notated H_0 , is a claim that there really is “no effect” or “no difference”, or represents the status quo.

An *alternative hypothesis*, notated H_A , is the claim for which we seek discernible evidence, and is established by observing evidence that contradicts the null hypothesis and supports the alternative hypothesis.

In the first step of a hypothesis test, we must state the hypotheses. The hypotheses consist of the *null hypothesis* and the *alternative hypothesis*. The null hypothesis, denoted H_0 , often represents a skeptical perspective or a claim of “no difference” or “no effect.” It assumes that any observed pattern in the data is due to chance alone. The alternative hypothesis, denoted H_A , represents

the claim under investigation, typically that there is a real difference, effect, or relationship. The alternative hypothesis is usually the reason the research was conducted in the first place.

Statistical tests, and in particular hypothesis tests, behave in much of the same way as the US justice system. The US court system considers two possible claims about a defendant: they are either innocent or guilty. We can set up these claims in a hypothesis framework. The skeptical perspective (null hypothesis) is that the person is innocent until evidence is presented that convinces the jury the person is guilty (alternative hypothesis).

H_0 : The defendant is innocent.

H_A : The defendant is guilty.

Caution

Notice that if a jury finds a defendant *not guilty*, this does not necessarily mean the jury is confident in the person's innocence. They are simply not convinced of the alternative. This is also the case with hypothesis testing: *even if we fail to reject the null hypothesis, we do not accept the null hypothesis as truth*. Failing to find evidence in favor of the alternative is not the same as finding evidence that the null hypothesis is true.

The courtroom gives us one more useful idea. A jury is instructed to convict only when the evidence establishes guilt *beyond a reasonable doubt* — a deliberately high bar, set so that the system rarely convicts an innocent person. A hypothesis test works the same way: we reject the null hypothesis only when the data would be genuinely *surprising* if the null were true. Later we will make “surprising” precise with a **significance level**, which is nothing more than a choice of where to set the bar for reasonable doubt. Keeping that bar high is what protects us from too often declaring an effect that isn't really there.

The shape of a hypothesis

Every test in this chapter begins the way the courtroom does — from skepticism. The **null hypothesis** plays the role of the presumption of innocence: it claims that *nothing is going on*, so the parameter sits at its “no-effect” value. For a single group that value is whatever is claimed or expected (p_0 or μ_0); for a comparison of two groups, “no effect” means the difference is **zero**. The **alternative hypothesis** is the claim the researcher actually set out to support, and it points in a direction the research question chooses — *different from*, *greater than*, or *less than* the null value.

That one idea is the whole story. Only the parameter changes from one setting to the next:

The question is about...	Null hypothesis (“no effect”)	Alternative (pick one direction)	A question of this kind
one proportion p	$p = p_0$	$p \neq p_0$, $p > p_0$, $p < p_0$	Do more than half of children pick the cartoon box? ($p_0 = 0.5$)
one mean μ	$\mu = \mu_0$	$\mu \neq \mu_0$, $\mu > \mu_0$, $\mu < \mu_0$	Do the bags really average the 16 oz on the label? ($\mu_0 = 16$)

The question is about...	Null hypothesis (“no effect”)	Alternative (pick one direction)	A question of this kind
a difference in proportions	$p_1 - p_2 = 0$	$p_1 - p_2 \neq 0, > 0, < 0$	Do two groups differ in who skips the video?
a difference in means	$\mu_1 - \mu_2 = 0$	$\mu_1 - \mu_2 \neq 0, > 0, < 0$	Is average income higher in eastern than western states?

The direction comes straight from the wording of the question. An alternative built with $>$ or $<$ is **one-sided** — the question asked about a particular direction (“greater,” “more than,” “faster”). An alternative built with $\neq$ is **two-sided** — the question only asks whether there is *any* difference at all (“different,” “not fair,” “changed”).

Class Example 7.1: Defining statistical hypotheses

For each of the following scenarios, state the null and alternative hypotheses with the relevant parameter(s).

- a) We are testing to see if the average income is different in eastern states than western states.

SOLUTION

- Parameters: μ_E and μ_W , the mean income in eastern and western states.
- $H_0 : \mu_E - \mu_W = 0$ (no difference in average income).
- $H_A : \mu_E - \mu_W \neq 0$ (two-sided — “different” gives no direction).

- b) We are testing to see if the average income is greater in eastern states than western states.

SOLUTION

- Parameters: μ_E and μ_W as above.
- $H_0 : \mu_E - \mu_W = 0$.
- $H_A : \mu_E - \mu_W > 0$ (one-sided — “greater” points east).

- c) We are testing to see if a particular coin is weighted (i.e., not fair).

SOLUTION

- Parameter: p , the true proportion of heads for this coin.
- $H_0 : p = 0.5$ (the coin is fair).
- $H_A : p \neq 0.5$ (two-sided — “not fair” could tip either way).

- d) We are testing to see if the average rent in La Crosse is less than \$850.

SOLUTION

1. Parameter: μ , the mean rent in La Crosse.
2. $H_0 : \mu = 850$.
3. $H_A : \mu < 850$ (one-sided — “less than”).

- e) We are testing to see if children have a preference for the cereal box with cartoons (above).

SOLUTION

1. Parameter: p , the proportion of children who choose the cartoon box.
2. $H_0 : p = 0.5$ (no preference).
3. $H_A : p > 0.5$ (one-sided — a preference *for* the cartoon box).

Case study: Opportunity cost

How rational and consistent is the behavior of the typical American college student? Here, we explore whether reminding students that money not spent now can be spent later causes them to be thriftier.

The participants were 150 students, all given a scenario about purchasing a video on sale for \$14.99. Half (the control group) were given these options:

- A. Buy this entertaining video.
- B. Not buy this entertaining video.

The other half (the treatment group) saw a slightly modified option (B):

- A. Buy this entertaining video.
- B. Not buy this entertaining video. Keep the \$14.99 for other purchases.

In this study, we define a “success” as a student who chooses *not* to buy the video and want to compare the proportion of students who do not buy the video in the control group compared to the treatment group.

Class Example 7.2: Opportunity cost

In the *opportunity cost* study, 150 college students each decided whether to buy a \$14.99 video. The 75 students in the **control** group chose between “buy” and “not buy,” while the 75 in the **treatment** group saw a reworded reminder — “not buy, *and keep the \$14.99 for other purchases.*” A “success” is a student who chooses *not* to buy the video. Using this study, answer the following questions:

- a) Is this an observational study or an experiment? How does the type of study impact what can be inferred from the results?

SOLUTION

1. It is an **experiment**: the researchers randomly assigned which version of option (B) each student saw.
2. Because the treatment was randomly assigned, the two groups are comparable, so a discernible difference can be attributed to the phrasing — we can make a **causal** claim, not just an association.

b) State the null and alternative hypotheses for this study.

SOLUTION

1. Parameters: p_T and p_C , the proportion who do **not** buy the video in the treatment and control groups.
2. $H_0 : p_T - p_C = 0$ (the reminder phrasing has no effect).
3. $H_A : p_T - p_C > 0$ (the reminder increases the proportion who choose not to buy).

The results are summarized below:

	Buy video	Not buy video	Total
Control	56	19	75
Treatment	41	34	75
Total	97	53	150

Class Example 7.3: Opportunity cost, revisited

Compute the difference in the sample proportions of students who choose not to buy the video.

SOLUTION

1. Control: $\hat{p}_C = 19/75 = 0.253$.
2. Treatment: $\hat{p}_T = 34/75 = 0.453$.
3. Difference: $\hat{p}_T - \hat{p}_C = 0.453 - 0.253 = 0.20$ (treatment group is 20 percentage points more likely to not buy).

Building a randomization distribution

What would it mean if the null hypothesis were true? It would mean that each student would make the decision on whether to buy the video without regard to the change in the second option, (B). The 97 purchases and 53 non-purchases were determined by factors other than the verbiage of the options, and the difference we observed was just bad luck in how the files happened to be assigned.

Suppose the students' decision were independent of the phrasing of the choices. Then, if we conducted the experiment again with a different random assignment of choices to the students, differences in the proportions of the choices would be based only on random fluctuation in decisions. We can perform this *randomization*, which simulates what would have happened if the students' decisions had been independent of the phrasing but we had distributed the phrases differently.

The card-shuffling metaphor

ACTIVITY

Think of the data as 150 cards:

- 97 red cards (buy video)
- 53 white cards (not buy video)

In the actual study, 75 cards were dealt to the ‘control’ pile and 75 to the ‘treatment’ pile. Under the null hypothesis, the color of each card (whether to buy the video) has nothing to do with which pile it lands in.

To simulate what the null hypothesis look like, we:

1. *Thoroughly shuffle* all 150 cards together.
2. *Deal* 75 into a ‘control’ pile and 75 into a ‘treatment’ pile
3. *Count* the number of white (not buy video) in each pile and compute the difference in proportions.

This gives us one simulated difference *under the null hypothesis*, a difference that arose purely from the random shuffle, not from the phrasing.

Many simulations

A *randomization distribution* approximates a sampling distribution from a population where the null hypothesis is true.

One simulation is not enough. We need to repeat the shuffle-and-deal process many times to build up a picture of what chance alone looks like. Using a computer, we can do this hundreds or thousands of times creating a *randomization distribution* showing how likely different statistics are to occur if H_0 is true.

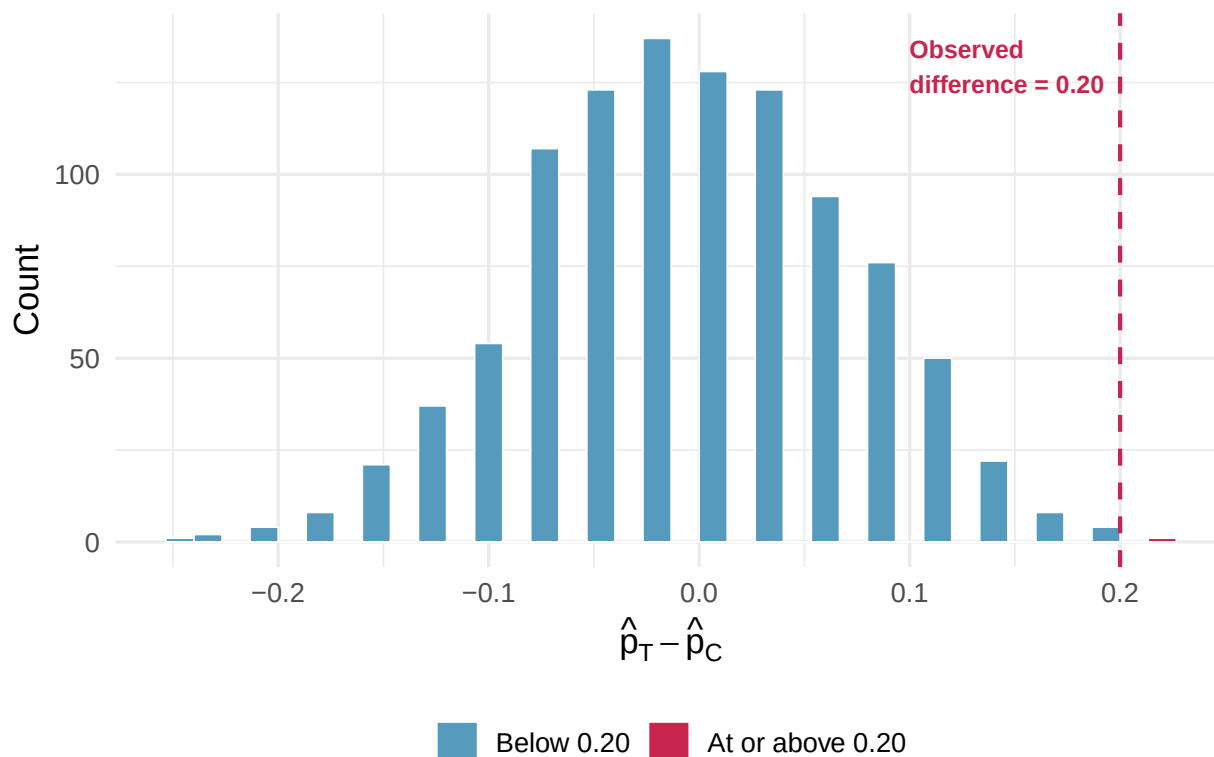


Figure 2: A histogram of 1,000 simulated differences produced under the null hypothesis. Differences at least as large as the observed 0.20 are highlighted in red.

Under the null hypothesis, we would observe a difference of at least +20% about 0.6% of the time. That is very rare! We conclude the data provide strong evidence there is a treatment effect: reminding students that they could spend money later lowers their chance of making a purchase. Because this is an experiment (with random assignment), we can make a causal claim.

P-values

The *p-value* of the sample data in a statistical test is the probability of obtaining a sample statistic as extreme as (or more extreme than) the observed sample when the null hypothesis is *true*.

When we claim that we have evidence of our alternative hypothesis, the underlying question we are asking is

How unusual is it to see a sample statistics as extreme as the sample statistic we observed, *if H_0 is true*?

In statistics, we use the *p-value* to answer this question. The *p-value* is the proportion of the randomization distribution that is as extreme (or more extreme than) over observed statistic. Since the randomization distribution is built assuming the null hypothesis is true, the *p-value* is also defined as the probability of observing data at least as favorable to the alternative hypothesis as our current dataset, if the null hypothesis were true. We typically use a summary statistic of the data, such as a difference in the proportion, to compute the *p-value*. We call this summary value the *sample statistic* — here, the observed difference in proportions. (Later, in the theory-based

chapters, we will standardize a sample statistic into a z - or t -score; that standardized value is what we will call the *test statistic*. For randomization tests we work directly with the sample statistic.)

If the sample statistic lies in the tails of the distribution, we have strong evidence against H_0 in support of H_A . The smaller the p -value, the more evidence we have to reject the null hypothesis.

Randomization Distributions and p -values

In order to correctly estimate the p -value from a randomization distribution, we need to keep our alternative hypothesis in mind!

To find the p -value using a randomization distribution, we:

- *For a one-sided alternative hypothesis:* Find the proportion of randomization samples that equal or exceed the original statistic in the direction (tail) indicated by the alternative hypothesis.
- *For a two-sided alternative hypothesis:* Find the proportion of randomization samples in the smaller tail at or beyond the original statistic and then double the proportion to account for the other tail.

Class Example 7.4: Estimating a p -value from a randomization distribution

The figure below shows a randomization distribution for testing $H_0 : p = 0.3$ vs $H_A : p < 0.3$. In each case, use the distribution to decide which value is closer to the p -value for the observed sample proportion. There are 1000 randomizations given in the figure below.

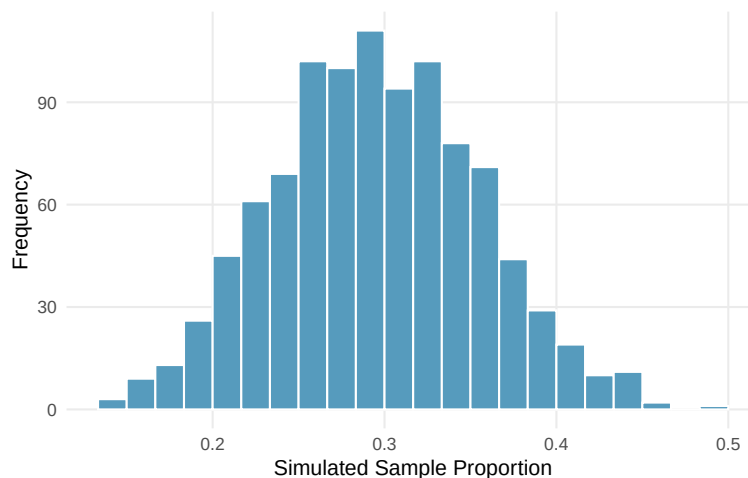


Figure 3: Randomization distribution with 1000 randomizations

a) Why is the distribution centered at 0.3?

SOLUTION

1. The randomization distribution is built *assuming* H_0 is true — that the true proportion is 0.3.
2. Each simulated sample proportion is generated with $p = 0.3$, so they pile up around 0.3; the spread shows the sampling variability we get by chance alone.

b) The p -value for $\hat{p} = 0.15$ is closest to: 0.04 or 0.40?

SOLUTION

1. $H_A : p < 0.3$, so the p -value is the *lower-tail* area — the fraction of simulations at or below 0.15.
2. 0.15 is far into the left tail, where very few simulations land, so the p -value is small: closest to **0.04**.

c) Which proportion, $\hat{p} = 0.20$ or $\hat{p} = 0.10$, provides the strongest evidence to suggest that H_0 is not true? Explain.

SOLUTION

1. Smaller p -value = stronger evidence against H_0 .
2. $\hat{p} = 0.10$ is farther into the lower tail than $\hat{p} = 0.20$, so fewer simulations are that extreme — its p -value is smaller.
3. So $\hat{p} = 0.10$ provides the strongest evidence that H_0 is not true.

Class Example 7.5: Estimating a p -value from a randomization distribution...continued

The figure below shows a randomization distribution for testing $H_0 : \mu_1 - \mu_2 = 0$ vs $H_A : \mu_1 - \mu_2 \neq 0$. The statistic used for each sample is $D = \bar{x}_1 - \bar{x}_2$. In each case, use the distribution to decide which value is closer to the p -value for the observed sample difference in averages. There are 1000 randomizations given in the figure below.

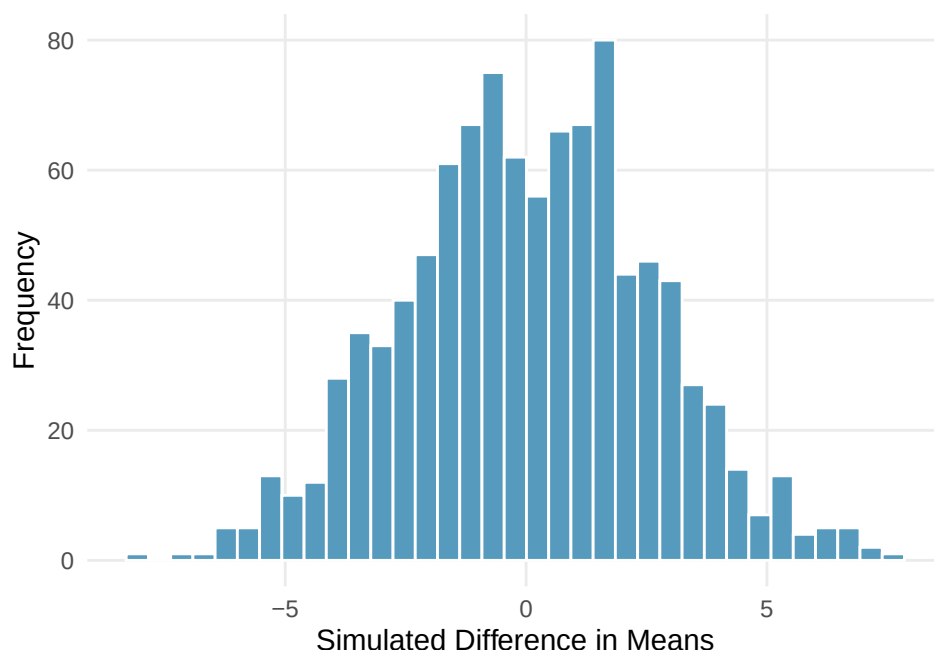


Figure 4: Randomization distribution with 1000 randomizations

a) The p -value for $D = \bar{x}_1 - \bar{x}_2 = -2.9$ is closest to: 0.01 or 0.25?

SOLUTION

1. $H_A : \mu_1 - \mu_2 \neq 0$ (two-sided), so count simulations at least as far from 0 as 2.9 in either tail.
2. -2.9 is only modestly out from the center, so a fair share of simulations are that extreme: closest to **0.25**.

b) The p -value for $D = \bar{x}_1 - \bar{x}_2 = 1.2$ is closest to: 0.30 or 0.60?

SOLUTION

1. 1.2 is very close to the center of the distribution, so a large majority of simulations are at least that far from 0.
2. The two-sided p -value is large: closest to **0.60**.

c) Of the two values of D above, which provides strongest evidence to suggest that H_0 is not true? Explain.

SOLUTION

1. $D = -2.9$ has the smaller p -value (≈ 0.25 vs. ≈ 0.60), so it sits farther in the tail.
2. So $D = -2.9$ gives the stronger evidence against H_0 — though with $p \approx 0.25$ neither result is discernible at $\alpha = 0.05$.

Class Example 7.6: Migraine Acupuncture Trial

Can acupuncture relieve migraine pain? Allais et al. (2011) conducted a randomized controlled study where 89 individuals who suffered from migraines were randomly assigned to receive either acupuncture (treatment, $n = 43$) or placebo acupuncture (control, $n = 46$). The placebo acupuncture involved placing needles at non-acupuncture-point locations. Researchers recorded whether each patient was pain-free 24 hours after treatment. The full data are in the data set *Migraine Acupuncture Trial* on StatLens.

a) What are the null and alternative hypotheses for this study?

SOLUTION

1. Parameters: p_T and p_C , the proportion pain-free after acupuncture (treatment) and sham (control).
2. $H_0 : p_T - p_C = 0$ (acupuncture is no better than the placebo).
3. $H_A : p_T - p_C > 0$ (acupuncture increases the proportion who are pain-free).

b) What was the sample statistic for these data?

SOLUTION

1. Treatment: $\hat{p}_T = 10/43 = 0.233$. Control: $\hat{p}_C = 2/46 = 0.043$.
2. Statistic: $\hat{p}_T - \hat{p}_C = 0.233 - 0.043 = 0.19$ (about 19 percentage points more pain-free in the acupuncture group).

c) What is the probability that the sample results (or those more extreme) occurred by random chance, assuming that the acupuncture had no effect?

SOLUTION

1. This probability is the p -value. Build the randomization distribution in StatLens by shuffling the 12 pain-free labels among all 89 patients and re-splitting into groups of 43 and 46; record how often the shuffled difference reaches 0.19 or more.
2. Very few shuffles reach 0.19: the one-sided p -value is about **0.01**. It would be rare to see a gap this large by chance alone.

d) How would your answer to c) change if the alternative hypothesis was two-sided?

SOLUTION

1. A two-sided test counts extremes in *both* tails, so we double the one-sided value: $p \approx 0.02$.
2. The evidence is a bit weaker, but $0.02 < 0.05$, so the conclusion (discernible effect) is unchanged.

Class Example 7.7: Calorie Counts

Is the average amount of kilocalories from a major U.S. fast food chain menu item more than 525 kilocalories? Calorie counts for 515 menu items from eight major fast food chains were collected. The full data are in the data set *Fast Food Calories* on StatLens.

a) State the null and alternative hypotheses

SOLUTION

1. Parameter: μ , the mean calorie count for fast food menu items.
2. $H_0 : \mu = 525$.
3. $H_A : \mu > 525$ (one-sided — “more than 525”).

b) What is the sample statistic?

SOLUTION

1. The sample mean: $\bar{x} \approx 531$ kilocalories (from the 515 menu items).

c) To create the randomization distribution, what do we have to assume?

SOLUTION

1. We assume H_0 is true — that the mean really is 525.
2. In StatLens we impose this by *shifting* the data so its mean sits at 525, then re-sampling to see how much the sample mean bounces around when $\mu = 525$. The distribution is therefore centered at 525.

d) What is the p -value for this study?

SOLUTION

1. The p -value is the fraction of that null distribution at or above the observed $\bar{x} \approx 531$.
2. 531 sits well inside the bulk of the distribution, so the p -value is large: about **0.32**. The data do **not** provide discernible evidence that the mean exceeds 525 kcal.

Class Example 7.8: Cereal study revisited

Recall the motivating cereal study. We were trying to determine if children had a preference for the cereal in the box with the cartoon. In the study 52 out of 80 children chose the box with cartoons. Use StatLens to find the p -value for this scenario.

SOLUTION

1. $H_0 : p = 0.5$ vs. $H_A : p > 0.5$; observed $\hat{p} = 52/80 = 0.65$.
2. In StatLens, simulate many samples of 80 children with $p = 0.5$ and find the fraction with $\hat{p} \geq 0.65$ (52 or more choosing cartoons).
3. That upper tail is tiny: the p -value is about **0.005**. Strong evidence that children do prefer the cartoon box.

Interpreting the p -value

- A *small p -value* (e.g., below 0.05) means that the observed result would be rare if the null hypothesis were true. This is evidence *against* the null hypothesis.
- A *large p -value* (e.g., above 0.10) means that the observed result is not particularly unusual under the null hypothesis. The data do not provide strong evidence against the null hypothesis.
- The p -value is *not* the probability that the null hypothesis is true. It is the probability of the observed data (or more extreme data) *given that* the null hypothesis is true.

Caution

A common mistake is to interpret the p -value as the probability that the null hypothesis is true (e.g., “there is a 2% chance that sex has no effect on promotions”). This is incorrect. The p -value tells us how *surprising* the data would be *if* the null hypothesis were true, not how likely the null hypothesis is to be true.

Statistical discernibility

Statistical tests begin under the assumption that the null hypothesis, H_0 , is true. In order to reject H_0 , sufficient evidence to its contrary must be collected – just like in order to convict a defendant, sufficient evidence must be presented to the judge and/or jury. What exactly is sufficient evidence? In our courtroom example, evidence must be provided so that decisions can be made “beyond a reasonable doubt.” In statistical tests, the data results must be unrealistic assuming H_0 is true. This means that we have found evidence to support H_A .

Decision framework

When we make a formal decision, we have two options:

- Reject H_0
- Do not reject H_0

For hypothesis tests, there are two conclusions:

- We *reject* H_0 when the sample statistic is extreme assuming H_0 is true.
- We *do not reject* H_0 when the sample statistic is not too extreme assuming H_0 is true.

If we reject the null hypothesis, we say our finding is *statistically discernible*.

If we reject the null hypothesis, we say our finding is *statistically discernible*. Not rejecting the null hypothesis is *not* the same thing as accepting the null hypothesis. The null hypothesis is *never accepted* because the hypothesis test procedure is not designed to *rule in* H_0 , but only to rule it out. That is, evidence is never collected with the idea of favoring or proving the null hypothesis. Sample data are collected for the purpose of looking for evidence *against* H_0 . To arrive at the correct conclusion, the p -value must be small enough to be considered unusual.

What “statistically discernible” does and does not mean

“Statistically discernible” means the p -value fell below the chosen discernibility level. It does *not* necessarily mean the effect is large or practically important. A very large sample can detect a tiny difference as “statistically discernible” even if the difference is too small to matter in practice.

Conversely, “not statistically discernible” does not mean there is no effect. It may simply mean the sample was too small to detect the effect.

How small is small enough?

We use the *discernibility level*, α , to decide the cutoff for our p -value to be discernible.

We discussed how small p -values mean that a statistic is unlikely if H_0 is true. Now we need to think about some rules to determine when our p -value is small enough to say something is unusual assuming the null hypothesis is true.

Common values for α : 0.05, 0.01, 0.10

We reject H_0 when $p\text{-value} < \alpha$

We do not reject H_0 when $p\text{-value} \geq \alpha$

The *discernibility level*, notated α , is the point at which a p -value is considered unusual. Some commonly used significance levels are $\alpha = 0.05$, $\alpha = 0.01$, or $\alpha = 0.10$. Any p -value that is smaller than the significance level is considered unusual. So, to use a p -value to determine whether or not the results are discernible, you should use the following guidelines:

- When the $p\text{-value} < \alpha$, reject H_0 .
- When the $p\text{-value} \geq \alpha$, do not reject H_0 .

Class Example 7.9: Red wine and weight loss

Resveratrol, an ingredient in red wine and grapes, has been shown to promote weight loss in rodents. A recent study (*Science Daily*, 2010) investigates whether the same phenomenon holds true in primates. A sample of six lemurs had their resting metabolic rate, body mass gain, food intake, and locomotor activity measured for one week prior to resveratrol supplementation and then the four indicates were measured again after treatment. For each of the following, use the p -value to make the appropriate conclusion (reject or do not reject) for the hypothesis tests. Use $\alpha = 0.05$ for each of your decisions.

- a) In a test to see if mean resting metabolic rate is higher after treatment (p -value of 0.013).

SOLUTION

1. $p = 0.013 < 0.05 = \alpha$, so **reject** H_0 .

- b) In a test to see if mean body mass gain is lower after treatment (p -value of 0.007).

SOLUTION

1. $p = 0.007 < 0.05 = \alpha$, so **reject** H_0 .

- c) In a test to see if mean locomotor activity is affected by treatment (p -value of 0.980).

SOLUTION

1. $p = 0.980 \geq 0.05 = \alpha$, so **do not reject** H_0 .

Interpreting your results

After you have made a decision to reject H_0 or not to reject H_0 , you need to interpret your conclusion in the context of the study. We always write our interpretation in the context of H_A .

A framework for interpreting your decision is as follows (fill in the blanks with the context of the study):

At $\alpha =$ (discernibility level), we (do/do not) have discernible evidence that (alternative hypothesis in words).

Class Example 7.10: Red wine and weight loss revisited

For each of the following, provide an interpretation of the conclusion in the context of the original study.

- a) In a test to see if mean resting metabolic rate is higher after treatment (p -value of 0.013).

SOLUTION

1. At $\alpha = 0.05$, we **do** have discernible evidence that the mean resting metabolic rate is higher after resveratrol treatment.

- b) In a test to see if mean body mass gain is lower after treatment (p -value of 0.007).

SOLUTION

1. At $\alpha = 0.05$, we **do** have discernible evidence that the mean body mass gain is lower after treatment.

c) In a test to see if mean locomotor activity is affected by treatment (p -value of 0.980).

SOLUTION

1. At $\alpha = 0.05$, we **do not** have discernible evidence that mean locomotor activity is affected by treatment.

Class Example 7.11: Migraine Acupuncture Trial revisited

Can acupuncture relieve migraine pain? Using your decision from the Acupuncture Trial example, interpret your results in the context of the problem.

SOLUTION

1. The p -value was about $0.01 < 0.05$, so we reject H_0 .
2. At $\alpha = 0.05$, we have discernible evidence that acupuncture increases the proportion of migraine patients who are pain-free 24 hours after treatment.
3. Because patients were randomly assigned, this supports a *causal* conclusion: the acupuncture, not chance, explains the higher pain-free rate.

Class Example 7.12: Calorie count revisited

Is the average amount of kilocalories from a major U.S. fast food chain menu item more than 525 kilocalories? Using your decision from the Calorie Count example, interpret your results in the context of the problem.

SOLUTION

1. The p -value was about $0.32 \geq 0.05$, so we do not reject H_0 .
2. At $\alpha = 0.05$, we do **not** have discernible evidence that the mean calorie count of fast food menu items exceeds 525 kilocalories.
3. Note this does not prove the mean equals 525 — only that the data are consistent with it.

Class Example 7.13: Cereal study revisited

Recall the motivating cereal study. Interpret the results of the study in the context of the problem.

SOLUTION

1. The p -value was about $0.005 < 0.05$, so we reject H_0 .
2. At $\alpha = 0.05$, we have discernible evidence that more than half of children prefer the cereal box with cartoons.

Type I and Type II errors

A *Type I error* occurs when we reject H_0 , but it is true (e.g., convict an innocent person).

A *Type II error* occurs when we do not reject H_0 , but it is false (e.g., let a guilty person go free).

Every hypothesis test ends with one of two decisions: reject H_0 or fail to reject H_0 . And the truth is one of two realities: H_0 is true or H_A is true. Combining these gives four scenarios:

Table 3: Four possible outcomes of a hypothesis test.

	H_0 is true	H_A is true
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

Two of the four outcomes are correct decisions:

- Rejecting H_0 when H_A is true: we correctly detected the real effect.
- Failing to reject H_0 when H_0 is true: we correctly avoided a false alarm.

The other two outcomes are errors:

- *Type I error* (upper-left): We rejected H_0 but it was actually true. False alarm.
- *Type II error* (lower-right): We failed to reject H_0 but H_A was actually true. Missed detection.

Class Example 7.14: Red wine and weight loss revisited

For each of the following, identify which kind of error could have occurred (Type I or Type II) using $\alpha = 0.05$.

- a) In a test to see if mean resting metabolic rate is higher after treatment (p -value of 0.013).

SOLUTION

1. $p = 0.013 < 0.05$, so we reject H_0 . When we reject, the only possible error is a **Type I error** (rejecting a true H_0).

- b) In a test to see if mean body mass gain is lower after treatment (p -value of 0.007).

SOLUTION

1. $p = 0.007 < 0.05$, so we reject $H_0 \rightarrow$ a possible **Type I error**.

- c) In a test to see if mean locomotor activity is affected by treatment (p -value of 0.980).

SOLUTION

1. $p = 0.980 \geq 0.05$, so we do not reject H_0 . When we fail to reject, the only possible error is a **Type II error** (failing to reject a false H_0).

What can cause these errors?

The two types of errors that can be made both involve the hypothesis test leading to the wrong conclusion about the null hypothesis. This can happen because the process of random sampling does not give complete or perfect information about the population, and it is always possible to draw a rare sample just by chance. Recall that we have two options when making statistical inferences: *reject H_0* or *do not reject H_0* . It is important to carefully consider what these decisions could mean.

- Decision: Reject H_0 . The following outcomes are possible:
 1. You are making the correct decision; H_0 is not true.
 2. You are making a Type I error; the sample statistic was so far from the hypothesized value just by chance.
 3. Your sampling process or experiment was poor; the sample statistic is biased.
- Decision: Do not reject H_0
 1. You are making the correct decision; H_0 is true.
 2. You are making a Type II error; the sample statistic was close to the hypothesized value just by chance.
 3. Your sampling process or experiment was poor; the sample statistic is biased.

Balancing out the errors

The discernibility level, α , is the probability we make a Type I error.

The discernibility level, α , is the probability we make a Type I error. If we can choose the discernibility level and it gives the chance of an error being made, why not always make the discernibility level really, really low? This is because there are trade-offs between controlling for Type I and Type II errors, and balancing the risk of these errors is important.

The *power* of a test tells us how likely we are to reject H_0 when we should.

1. Setting α to be very small makes it difficult to make a Type I error, but it also makes it easier to make a Type II error. Conversely, setting α to be large increases the making a Type I error, but it also reduces the risk of making a Type II error.
2. Since a Type I error is usually thought to be more serious than a Type II error, we typically use a low α = Type I error probabilities like 0.05, 0.1, or 0.01
3. We can reduce the chance of a Type II error without impacting the risk of a Type I error by increasing the sample size.
4. Statistical *power* tells us how likely we are to reject H_0 when we should (so we want *higher* power!), and it is related to the risk of a Type II error by $\text{power} = 1 - \text{chance of Type II error}$.
5. Calculating the probability of Type II error and power are fairly complicated for most cases, so we won't discuss that in this class. But understanding the relationships between these errors is important.

The *power* of a test tells us how likely we are to reject H_0 when we should

Class Example 4.15: Errors in a drug study

A pharmaceutical company is researching a new anti-anxiety drug. Marketing a new drug is costly, and the company only makes a profit when the drug is effective for over 80% of patients.

A random sample of 60 participants are placed on the drug and the number of participants that respond favorably to the drug will be recorded.

- a) Give the null and alternative hypothesis that you would use for determining if the drug should be marketed.

SOLUTION

1. Parameter: p , the proportion of patients who respond favorably.
2. $H_0 : p = 0.80$ (the drug works for exactly the break-even 80%).
3. $H_A : p > 0.80$ (worth marketing — effective for more than 80%). The burden of proof is on showing the drug clears 80%.

- b) Describe in words what a Type I error would mean in the context of this study.

SOLUTION

1. A Type I error = rejecting a true H_0 : concluding the drug helps *more* than 80% of patients (and marketing it) when in truth it does not.
2. Consequence: the company spends money marketing a drug that will not be profitable.

- c) Describe in words what a Type II error would mean in the context of this study.

SOLUTION

1. A Type II error = failing to reject a false H_0 : concluding there is not enough evidence to market the drug when it truly does help more than 80%.
2. Consequence: the company passes on a drug that would have been effective and profitable.

- d) Suppose the significance level is set at $\alpha = .05$, but the researchers find that this produces a Type II error rate of 0.4 and this is too high for their liking. List two ways they could reduce the Type II error rate.

SOLUTION

1. **Increase the sample size** (test more than 60 patients) — the surest way to lower the Type II rate without raising the Type I rate.
2. **Increase the significance level α** (say to 0.10) — makes rejecting H_0 easier, lowering the Type II rate (at the cost of a higher Type I rate).

- e) Given a Type II error rate of 0.4, what is the power of the test?

SOLUTION

1. Power = $1 - P(\text{Type II error}) = 1 - 0.4 = 0.6$.

The randomization test procedure: a summary

We can summarize the randomization test procedure as follows:

1. *Frame the research question in terms of hypotheses.* The null hypothesis (H_0) usually represents no difference or no effect. The alternative hypothesis (H_A) represents the claim of

interest.

2. *Collect data.* If the research question focuses on associations but not causation, use an observational study. If you want to establish a causal connection, use an experiment with random assignment.
3. *Model what would happen under the null hypothesis.* Shuffle (randomize) the data to break any association between the explanatory and response variables. Compute the sample statistic for each shuffle. Repeat many times to build the *randomization distribution* (also called the null distribution).
4. *Compute the p-value.* Determine what proportion of the randomization distribution is at least as extreme as the observed sample statistic.
5. *Draw a conclusion.* If the p-value is small (less than α), reject H_0 and conclude there is evidence for H_A . If the p-value is not small, fail to reject H_0 . Write the conclusion in context, in plain language.

Summary

- A *hypothesis test* evaluates two competing claims: the null hypothesis (H_0), which represents no effect or no difference, and the alternative hypothesis (H_A), which represents the claim under investigation.
- A *randomization test* (or permutation test) assesses the null hypothesis by shuffling the data to simulate what would happen if there were truly no effect. The collection of simulated sample statistics forms the *randomization distribution*.
- The *sample statistic* is the summary value computed from the data (e.g., a difference in proportions) that we locate on the randomization distribution to evaluate the hypotheses.
- The *p-value* is the probability of observing a sample statistic at least as extreme as the one observed, assuming the null hypothesis is true. It quantifies the strength of the evidence against H_0 .
- We reject H_0 when the p-value is less than the *discernibility level* α (commonly 0.05). This is called a *statistically discernible* result.
- A *one-sided test* looks for effects in a specified direction; a *two-sided test* looks for effects in either direction.
- Failing to reject H_0 is not the same as proving H_0 is true.
- Every hypothesis test ends in one of four outcomes, organized by the *decision table*: two correct decisions and two errors.
- A *Type I error* rejects a true H_0 (a false alarm); its probability is the discernibility level α , which we choose.
- A *Type II error* fails to reject H_0 when H_A is true (a missed detection); which we do not directly control.
- Statistical discernibility does not imply practical importance.